

SMC Fabric — Register Interface

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All fabric configuration registers are accessed through the local crossbar's internal CSR sub-crossbar. Software uses the same AXI4 path as all other local-space accesses; the fabric decodes the address range and dispatches to the appropriate internal handler.

All offsets are relative to `LOCAL_BASE` (default `0xC000_0000`).

In the LT model, `b_transport` writes to the remap and filter sockets update the fabric's internal tables immediately. There is no shadow-register latency.

1. Global Fabric CSRs

Served directly by the fabric's `b_transport` handler for the CPU control window at `LOCAL_BASE + 0x001_0000`.

Offsets/resets per `ocah_smc_top_reg.svh` (`SMC_CPU_CTRL_*`).

Offset	Register	Access	Reset	Description
<code>0x10040</code>	<code>GLOBAL_BASE</code>	RW	<code>0x4000_0000</code>	Global-alias base. A write updates the routing parameter and triggers DMI invalidation on all initiator sockets.
<code>0x10048</code>	<code>LOCAL_BASE</code>	RO	<code>0xC000_0000</code>	Local aperture base. Returns the constructor-time <code>local_base_addr</code> value.
<code>0x10050</code>	<code>REGION_SIZE</code>	RW	<code>0x0200_0000</code>	Aperture window size (32 MiB default). Write triggers DMI invalidation.

2. Alias Remap Registers (`ALIAS_REMAP`)

Base: `LOCAL_BASE + 0x001_2000`

8 entries, 0x20 bytes each. Stride: `N × 0x20`.

Offset within entry	Field	Access	Description
<code>+0x00</code>	<code>REGION_START</code>	RW	Aligned start address of the region (4 KiB granularity)
<code>+0x08</code>	<code>REGION_END</code>	RW	Exclusive end address
<code>+0x10</code>	<code>REGION_OFFSET</code>	RW	Signed offset added to addresses within the region
<code>+0x18</code>	<code>REGION_ATTRS</code>	RW	[0] = <code>valid</code> — enable this entry; [1] = <code>cacheable</code> — propagated to AXI user field

LT side-effects on write: Rebuilds the internal `alias_regions_[8]` table. Calls `invalidate_direct_mem_ptr(0, UINT64_MAX)` on all initiator sockets.

Constants:

Name	Value	Description
Number of entries	8	
Address granularity	4 KiB (bits [11:0] of all address fields are zero)	

3. M-Mode Remap Registers (`MMODE_REMAP`)

Base: `LOCAL_BASE + 0x001_3000`

8 entries, 0x08 bytes each. Stride: $N \times 0x08$. Bit-exact image of `output_remap.rd1` / `output_remap.sv` (lines 52–61): each entry is a single 64-bit `REGION_ATTRS` register and there is **no valid bit** — every region is always active.

Offset within entry	Field	Access	Description
+0x00	<code>REGION_ATTRS.offset[31:0]</code>	RW	Low 32 bits of the 56-bit translation offset
+0x04	<code>REGION_ATTRS.offset[55:32]</code>	RW	High 24 bits (a 64-bit access at +0x00 reads/writes the whole field)

Behaviour: Bit-exact to `output_remap.sv`. For a transaction whose address falls within the M-mode window (`MMODE_REMAP_START ... +MMODE_REMAP_SIZE`, relative to `LOCAL_BASE` or `GLOBAL_BASE`):

```
adjusted = addr - window_base
idx       = adjusted[OUTPUT_REMAP_IDX_START +: clog2(8)] // bits [22:20]
out       = { offset[55:OUTPUT_REMAP_IDX_START], adjusted[OUTPUT_REMAP_IDX_START-1:0] }
```

i.e. the 8 regions tile the 8 MiB window in 1 MiB (`IdxStart = 20`) slots; the selected region's `offset[55:20]` replaces the upper address bits while the low 20 bits are preserved. The `smc_axi_extension.source_id` is overridden to `MMODE_SRC_ID`. (At reset `offset == 0`, matching the RTL default.)

LT side-effects on write: Updates `mmode_regions[idx].offset` (raw register image). Triggers DMI invalidation.

4. Xvisor Remap Registers (`XVISOR_REMAP`)

Base: `LOCAL_BASE + 0x001_4000`

8 entries, 0x08 bytes each. Same layout as M-mode remap — a single 64-bit `REGION_ATTRS.offset[55:0]` per entry, no valid bit.

Offset within entry	Field	Access	Description
+0x00	REGION_ATTRS.offset[31:0]	RW	Low 32 bits of the 56-bit translation offset
+0x04	REGION_ATTRS.offset[55:32]	RW	High 24 bits

Behaviour: Xvisor (H-mode) traffic in the Xvisor window is remapped using the same {offset[55:20], adjusted[19:0]} encoding as M-mode (see §3) and tagged with OTHERS_SRC_ID.

LT side-effects on write: Updates xvisor_regions_[idx].offset. Triggers DMI invalidation.

5. Inbound Filter Registers

Base: LOCAL_BASE + 0x001_5000 (SMC_INBOUND_FILTER_CTRL_0, decoded by the internal CSR crossbar). 16 filter entries, stride 0x20 (entry N at base + $N \times 0x20$). Each entry is a bit-exact image of filter_ctrl1.rdl — three 64-bit registers:

Offset within entry	Register	Access	Description
+0x00	FILTER_CONFIG	RW	read_en[0], write_en[1], addr_mode[4] (entry enable), allow_ns[8], data_bus_width[14:12] (RO = 3), src_id[19:16], group_id[23:20], allow_burst[24], locked[63]
+0x08	START_ADDR	RW	start_addr[55:0] — inclusive range start
+0x10	END_ADDR	RW	end_addr[55:0] — inclusive range end (reset 0x7)
+0x18	—	—	Reserved within the 0x20 stride

accesswidth = 64: a 64-bit access at the register offset reads/writes the whole field; 32-bit accesses see the low (+0x0) or high (+0x4) word.

Matching (bit-exact to axi_filter_wrap / traffic_filter): a filter *hits* when `addr_mode = 1` **and** the (granularity-snapped) address lies in `[start_addr, end_addr]` **and** the source-ID matches (`src_id = 0` ⇒ ignored, else exact match) **and** the NS bit matches exactly (`tx_ns == allow_ns`). Group-ID filtering is disabled in the SMC (`EnGroupIdFilter = 0`). The lowest-index hitting filter wins; the transaction is then permitted iff that filter's `read_en/write_en` (for the access direction) is set. Address granularity is 4 KiB when `allow_burst = 1`, otherwise 8 B (`data_bus_width`).

Default policy: Deny all (`BlockByDefault = 1` in `smc_input_fabric`) — when no filter hits, the transaction is blocked.

locked: write-one-to-set; once set, further writes to that entry's registers are ignored.

LT side-effects on write: Updates `inbound_filter_[16]` immediately. Bursts are not modelled in LT (single-beat payloads), so `allow_burst` only selects the match granularity.

6. Outbound Filter Registers

Base: `LOCAL_BASE + 0x001_6000` (`SMC_OUTBOUND_FILTER_CTRL_0`). Same entry layout and matching rules as the inbound filter.

Default policy: Allow all (`BlockByDefault = 0` in `smc_output_fabric`) — when no filter hits, the transaction passes.

LT side-effects on write: Updates `outbound_filter_[16]` immediately.

7. Register Access Semantics

Operation	LT behaviour
Any register read	Returns current model state; adds <code>reg_access_ns</code> to <code>delay</code>
Write <code>GLOBAL_BASE</code>	Updates <code>global_base_</code> routing parameter; triggers <code>invalidate_all_dmi()</code>
Write <code>REGION_SIZE</code>	Updates <code>region_size_</code> routing parameter; triggers <code>invalidate_all_dmi()</code>
Write <code>ALIAS_REMAP[N].*</code>	Rebuilds alias remap table; triggers <code>invalidate_all_dmi()</code>
Write <code>MMODE_REMAP[N].*</code>	Rebuilds M-mode remap table; triggers <code>invalidate_all_dmi()</code>
Write <code>XVISOR_REMAP[N].*</code>	Rebuilds Xvisor remap table; triggers <code>invalidate_all_dmi()</code>
Write inbound/outbound filter	Rebuilds the relevant filter table
<code>get_direct_mem_ptr</code>	Always returns <code>dmi_allowed = false</code>

8. Source-ID Constants

Values per `smc_pkg.sv` (`OTHERS_SRC_ID`, `SMC_SRC_ID`, `MMODE_SRC_ID`, `SEP_SRC_ID`).

Constant	Value	Applied on
<code>OTHERS_SRC_ID</code>	<code>0x0</code>	After Xvisor remap
<code>SMC_SRC_ID</code>	<code>0x3</code>	Plain outbound path (non-M-mode, non-Xvisor)
<code>MMODE_SRC_ID</code>	<code>0xC</code>	After M-mode remap
<code>SEP_SRC_ID</code>	<code>0xF</code>	SEP inbound traffic (not remapped in the LT model)